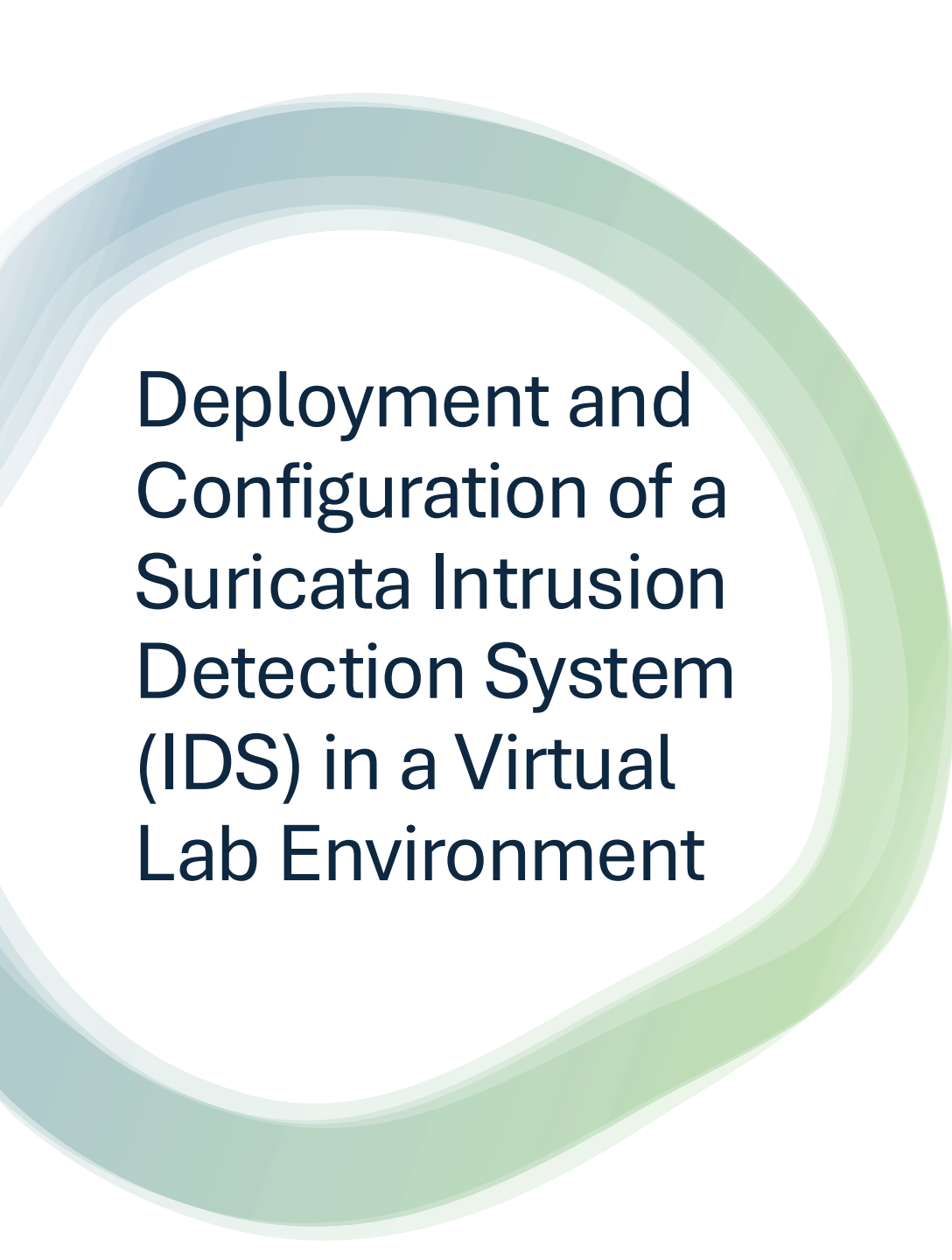
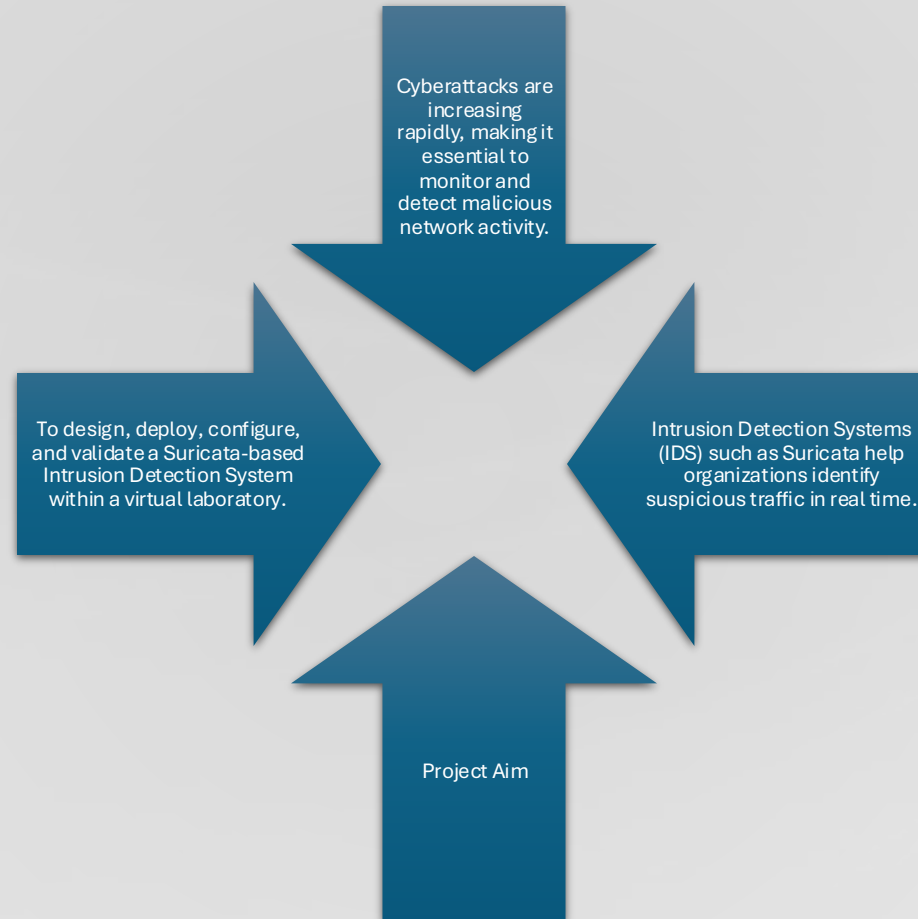




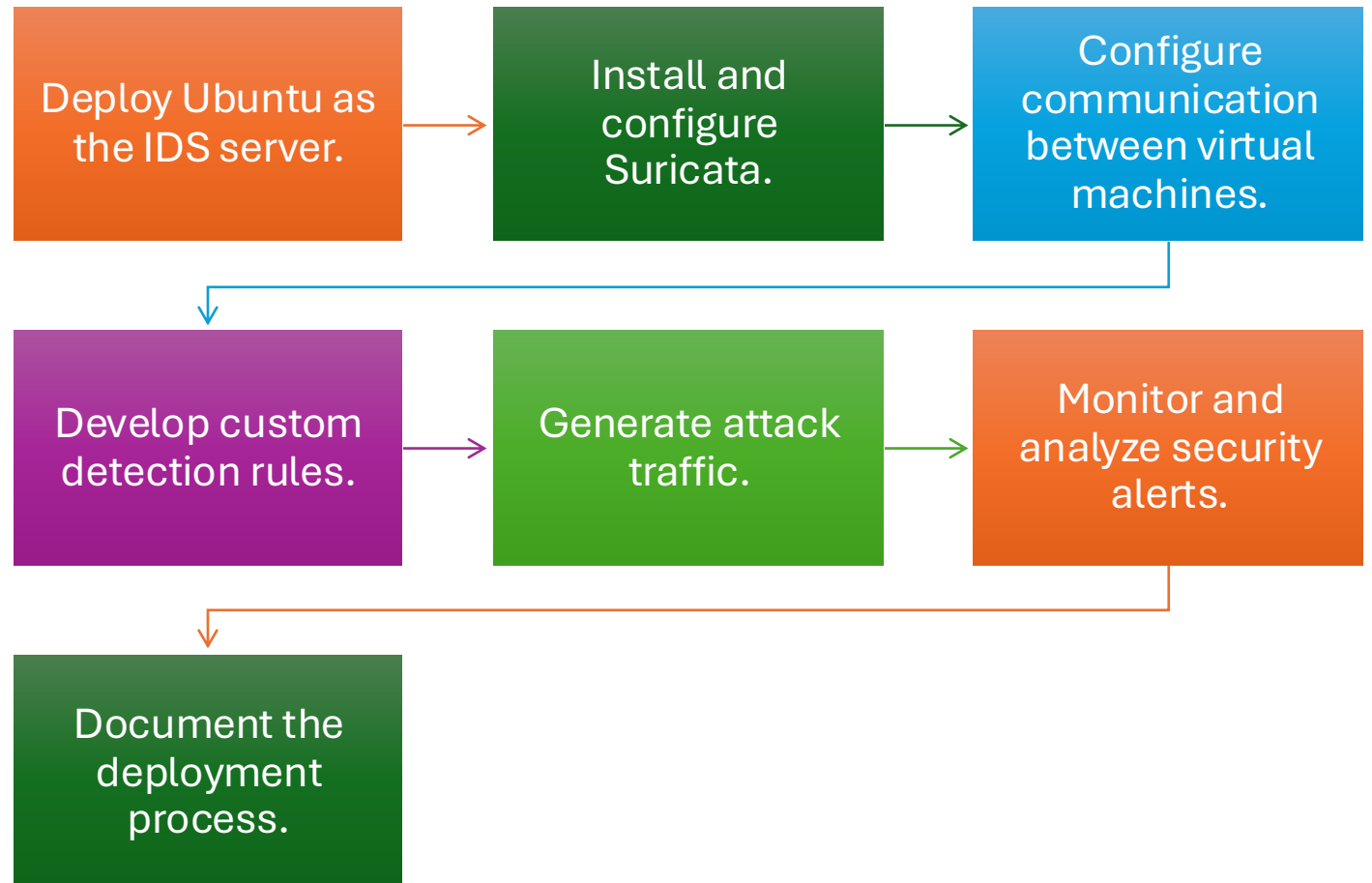
Deployment and Configuration of a Suricata Intrusion Detection System (IDS) in a Virtual Lab Environment

- **Presented by**
- **Blue team ,group 4**
- **Technology Stack**
 - Ubuntu Server 26.04 LTS
 - Kali Linux
 - Metasploitable2
 - Oracle VirtualBox
 - Suricata IDS

Project Background



Project Objectives





Lab Architecture

Virtual Environment

Machine		Role
Ubuntu Server		IDS Server
Kali Linux		Attacker
Metasploitable2		Target Server
Network		
Host-Only Adapter		
Kali Linux Ubuntu (Suricata IDS) Metasploitable2		



Tools Used

Tool	Purpose
Oracle VirtualBox	Virtualization
Ubuntu 26.04 LTS	IDS Server
Kali Linux	Attack Machine
Metasploitable2	Vulnerable Target
Suricata 8.x	Intrusion Detection
GitHub	Project Repository

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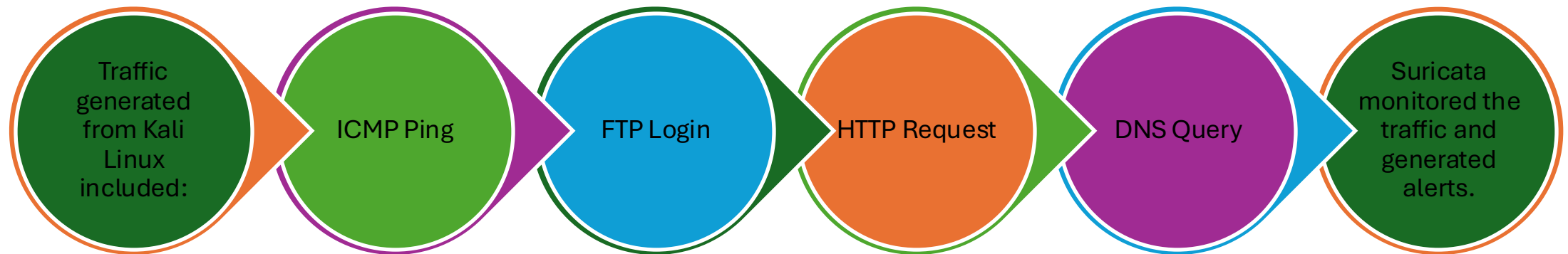
Installation and Configuration

- Completed Tasks
- Ubuntu installation
- System update
- Suricata installation
- Network interface configuration
- Rule configuration
- Service validation
- **Commands used**
- `sudo apt update`
- `sudo apt install suricata`
- `suricata --version`
- `sudo systemctl restart suricata`

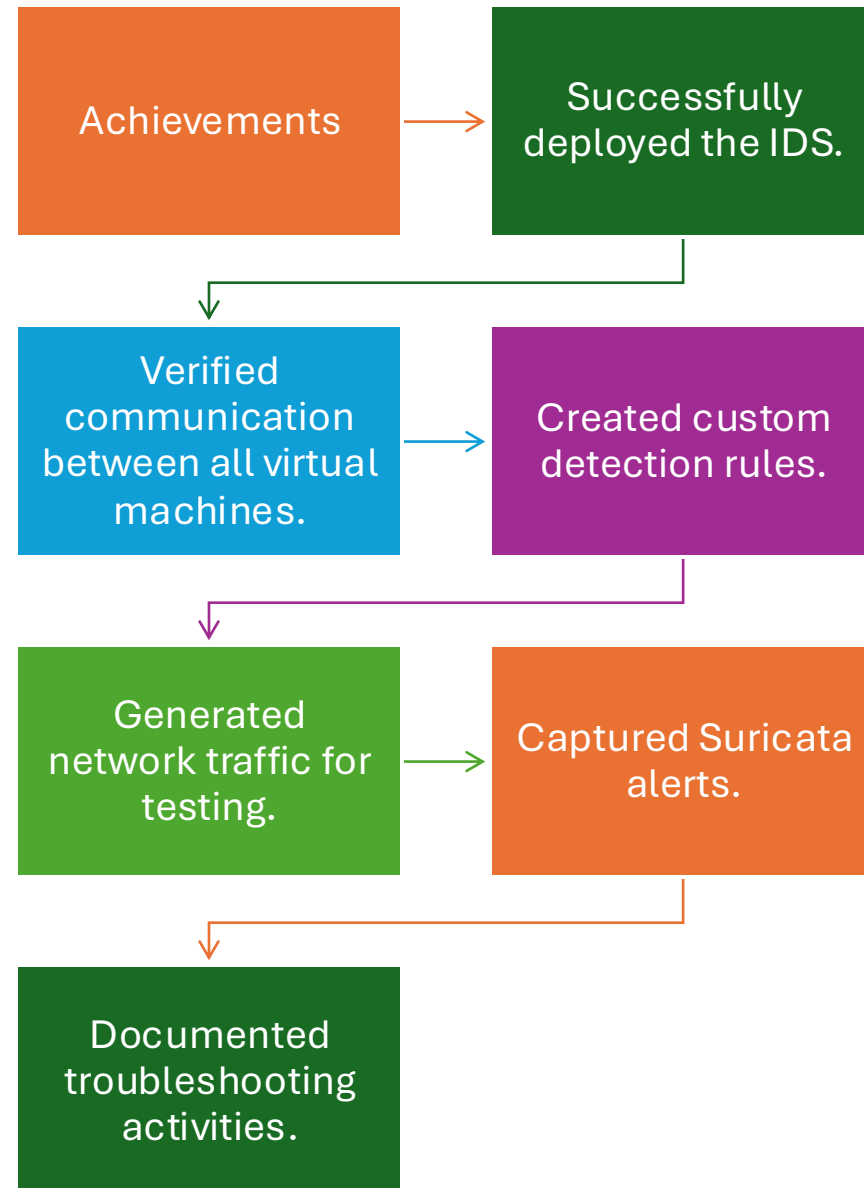
Custom Detection Rules

Rule	Purpose
ICMP	Detect Ping
HTTP	Detect Web Traffic
FTP	Detect FTP Connections
DNS	Detect DNS Queries
SSH	Detect SSH Connections
Telnet	Detect Telnet Sessions

Testing and Validation



Results



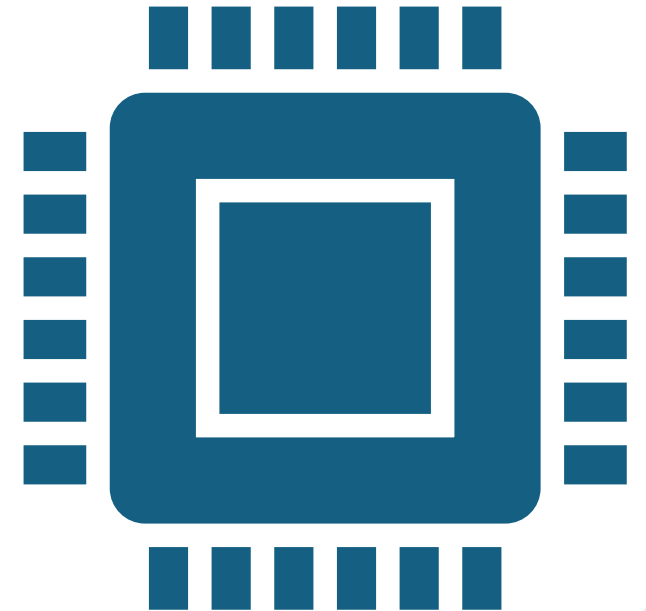
Challenges Encountered

- Virtual machine networking issues.
- YAML configuration errors.
- Rule loading issues.
- Keyboard copy/paste limitations.
- Service restart troubleshooting.
- Resolution
- Verified network interfaces.
- Corrected configuration files.
- Validated Suricata configuration.
- Restarted services.
- Retested IDS functionality.



Conclusion

- The project successfully demonstrated the deployment and configuration of a Suricata Intrusion Detection System in a virtual environment. The IDS detected simulated network activities and provided valuable insight into network monitoring, rule creation, and security analysis.



THANK YOU



THANK YOU



QUESTIONS
AND ANSWERS



PRESENTED BY



BLUE TEAM,
GROUP 4